Linear Algebra I — Week 4

Row Reduction, Solving Linear Systems & Bases from RREF

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Elementary row operations. (i) swap two rows $R_i \leftrightarrow R_j$; (ii) scale a row $R_i \to cR_i$ with $c \neq 0$; (iii) add a multiple of one row to another $R_i \to R_i + \lambda R_j$. They do not change the solution set of a system, and they can be undone, so they do not change the span of the rows.

REF and RREF. A matrix is in *row echelon form* (REF) if the zero rows are at the bottom and the first nonzero entry (the *leading entry*) of each row lies strictly to the right of the leading entry of the row above. It is in *reduced* row echelon form (RREF) if moreover every leading entry is 1 and is the only nonzero entry in its column. The RREF of a matrix is unique. A *pivot column* is a column containing a leading entry.

Solving $A\mathbf{x} = \mathbf{b}$. Reduce $\big(A \mid \mathbf{b}\big)$ to RREF. If a row reads $(0 \cdots 0 \mid 1)$ the system is *inconsistent*. Otherwise variables in pivot columns are *basic*, the others are *free*. No free variable: a unique solution. At least one free variable: infinitely many solutions, written as $\mathbf{p} + s\,\mathbf{v}_1 + t\,\mathbf{v}_2 + \cdots$ (parametric vector form).

Homogeneous systems. $A\mathbf{x} = \mathbf{0}$ is always consistent ($\mathbf{x} = \mathbf{0}$). Its solution set is a subspace, the span of the vectors multiplying the free parameters. If there are more unknowns than equations there is a free variable, hence a nonzero solution. If $\mathbf{p}$ solves $A\mathbf{x} = \mathbf{b}$, every solution is $\mathbf{p} + \mathbf{h}$ with $A\mathbf{h} = \mathbf{0}$.

Columns, rows and RREF. The columns of A at the pivot positions of its RREF form a basis of $\mathrm{col}(A)$ (take them from the original A, not from the RREF). The nonzero rows of the RREF form a basis of $\mathrm{row}(A)$. The entries of a non-pivot column of the RREF are the coefficients expressing that column of A through the pivot columns. Vectors $v_1, \ldots, v_k$ are independent exactly when the matrix with these columns has a pivot in every column.

Worked Examples

Example 1 · Reducing a matrix to RREF

Reduce $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ to RREF.

Solution.

1. Go column by column. Column 1: the top entry is already 1, so use it as the pivot and clear the entries below.
2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

3. Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

4. Apply $R_3 \to \frac{1}{2}R_3$:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

5. Apply $R_1 \to R_1 + R_3,\ R_2 \to R_2 - R_3$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

6. Every column has a pivot; the result is the identity matrix, which is in RREF.

Answer: RREF $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Example 2 · A system with a unique solution

Solve $\begin{cases} x + y = 3 \\ 2x - y = 0 \end{cases}$

Solution.

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array}\right)$$

3. Apply $R_2 \to \dfrac{1}{-3}R_2$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right)$$

4. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right)$$

5. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

6. Read each nonzero row as an equation for its basic variable:

$x = 1$

$y = 2$

7. Check by substituting $(x, y) = (1, 2)$ into the original equations: the left sides give $(3, 0)$, the right-hand sides ✓.

Answer: Unique solution: $(x, y) = (1, 2)$.

Example 3 · A system with infinitely many solutions

Solve $\begin{cases} x + y + z = 3 \\ 2x + y = 3 \end{cases}$

Solution.

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 1 & 0 & 3 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -1 & -2 & -3 \end{array}\right)$$

3. Apply $R_2 \to \dfrac{1}{-1}R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 3 \end{array}\right)$$

4. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 3 \end{array}\right)$$

5. Pivot columns are the columns of the basic variables: x, y. The remaining variable(s) z are *free*: call them $z = s$.

6. Read each nonzero row as an equation for its basic variable:

$x = s$

$y = 3 - 2s$

$z = s$

7. Write this as a vector: $\mathbf{x} = \begin{pmatrix}0\\3\\0\end{pmatrix} + s\begin{pmatrix}1\\-2\\1\end{pmatrix}$ (the parameters s may be any numbers).

8. Check with $s = 1$: $\mathbf{x} = \begin{pmatrix}1\\1\\1\end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(3, 3)$ ✓.

Answer: Infinitely many solutions: $\mathbf{x} = \begin{pmatrix}0\\3\\0\end{pmatrix} + s\begin{pmatrix}1\\-2\\1\end{pmatrix}$.

Example 4 · An inconsistent system

Solve $\begin{cases} x + y + 2z = 1 \\ 2x + 2y + 5z = 3 \\ x + y = 2 \end{cases}$

Solution.

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 2 & 5 & 3 \\ 1 & 1 & 0 & 2 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & 1 \end{array}\right)$$

3. Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 + 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right)$$

4. Row 3 reads $0 = 1$, a contradiction. No numbers x, y, z can satisfy it, so the system is *inconsistent*.

Answer: No solution (inconsistent).

Example 5 · A homogeneous system

Solve $\begin{cases} x + y + 2w = 0 \\ 2x + 2y + z + 5w = 0 \end{cases}$ and write the solution set as a span.

Solution.

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 2 & 0 \\ 2 & 2 & 1 & 5 & 0 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right)$$

3. Pivot columns are the columns of the basic variables: x, z. The remaining variable(s) y, w are *free*: call them $y = s$, $w = t$.
4. Read each nonzero row as an equation for its basic variable:

   $x = -s - 2t$

   $y = s$

   $z = -t$

   $w = t$

5. Write this as a vector: $\mathbf{x} = s\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}$ (the parameters s, t may be any numbers).
6. Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -3 \\ 1 \\ -1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

Answer: Infinitely many solutions: $\mathbf{x} = s\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}$. The solution set is span of the two vectors above.

Example 6 · Column space basis and column relations

For $A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 2 & 6 & 1 & 5 \\ 1 & 3 & 1 & 3 \end{pmatrix}$ find a basis of $\mathrm{col}(A)$ and express the other columns using it.

Solution.

1. Reduce A to RREF.
2. Apply $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$:

   $$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

3. Apply $R_3 \to R_3 - R_2$:

   $$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

4. Pivot columns: 1,3. By the pivot-column rule, a basis of $\mathrm{col}(A)$ is formed by columns 1,3 *of the original* A: $\left\{\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right\}$.
5. The non-pivot columns of the RREF give the coefficients: column 2 of R is $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, so $\mathrm{col}_2 =$

$3\,\mathrm{col}_1$; column 4 of R is $\begin{pmatrix}2\\1\end{pmatrix}$, so $\mathrm{col}_4 = 2\,\mathrm{col}_1 + 1\,\mathrm{col}_3$.

Answer: Basis $\{\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}\}$.

Example 7 · A system over $\mathbb{C}$

Solve $\begin{cases} x + iy = 1 + i \\ ix + 2y = i \end{cases}$ over $\mathbb{C}$.

Solution.

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ i & 2 & i \end{array}\right)$$

2. Apply $R_2 \to R_2 - iR_1$:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ 0 & 3 & 1 \end{array}\right)$$

3. Apply $R_2 \to \frac{1}{3}R_2$:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ 0 & 1 & \frac{1}{3} \end{array}\right)$$

4. Apply $R_1 \to R_1 - iR_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 1+\frac{2}{3}i \\ 0 & 1 & \frac{1}{3} \end{array}\right)$$

5. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

6. Read each nonzero row as an equation for its basic variable:

$$x = 1 + \frac{2}{3}i$$
$$y = \frac{1}{3}$$

7. Check by substituting $(x, y) = (1 + \frac{2}{3}i, \frac{1}{3})$ into the original equations: the left sides give $(1 + i, i)$, the right-hand sides ✓.

Answer: Unique solution: $(x, y) = (1 + \frac{2}{3}i, \frac{1}{3})$.

Example 8 · Proof: row reduction does not change the solutions

Prove: swapping two equations, scaling an equation by $c \neq 0$, or adding a multiple of one equation to another does not change the solution set.

Solution.

1. Swapping: the system is the same list of equations in a different order, and a vector solves a

list of equations regardless of order.

2. Scaling E_i by $c \neq 0$: if $E_i(\mathbf{x}) = \beta_i$ then $(cE_i)(\mathbf{x}) = c\beta_i$. Conversely, if $(cE_i)(\mathbf{x}) = c\beta_i$, multiply both sides by c^{-1} (possible because $c \neq 0$ in a field) to recover $E_i(\mathbf{x}) = \beta_i$.
3. Adding λE_j to E_i: if $\mathbf{x}$ solves both E_i, E_j it solves $E_i + \lambda E_j$. Conversely, if $\mathbf{x}$ solves E_j and $E_i + \lambda E_j$, then $E_i(\mathbf{x}) = (E_i + \lambda E_j)(\mathbf{x}) - \lambda E_j(\mathbf{x}) = \beta_i$.
4. Each operation is undone by an operation of the same kind, so solutions are preserved in both directions.

Answer: The three operations preserve the solution set, so a system and its RREF have the same solutions.

Practice Problems

Part A — Echelon Forms and Row Reduction

1. [Foundational]

Classify each matrix as: in RREF, in REF but not RREF, or neither. For every matrix that is not in RREF, name the condition it violates.

$$\text{(a)}\ \begin{pmatrix}1&0&2\\0&1&-1\\0&0&0\end{pmatrix}\quad \text{(b)}\ \begin{pmatrix}1&0&2\\0&1&0\\0&0&1\end{pmatrix}\quad \text{(c)}\ \begin{pmatrix}0&1&3\\1&0&2\\0&0&0\end{pmatrix}\quad \text{(d)}\ \begin{pmatrix}1&3&0&2\\0&0&1&-1\\0&0&0&0\end{pmatrix}$$

Step 1. Recall the two lists of conditions. *REF:* zero rows are at the bottom, and the first nonzero entry (the *leading entry*) of each row lies strictly to the right of the leading entry of the row above. *RREF:* REF, plus every leading entry equals 1, plus each leading entry is the only nonzero entry in its column.

Step 2. (a) Leading entries: row 1 in column 1, row 2 in column 2, row 3 is a zero row at the bottom. They move right, so it is REF. Both leading entries are 1. Column 1 has only the entry 1 (others 0). Column 2 has only the entry 1. So (a) is in RREF.

Step 3. (b) Leading entries sit in columns $1, 2, 3$, moving right, so it is REF. All leading entries are 1. But column 3 contains the leading entry 1 of row 3 and also the entry 2 above it in row 1. A leading entry must be the *only* nonzero entry in its column, so (b) is REF but *not* RREF.

Step 4. (c) Row 1 has its leading entry in column 2; row 2 has its leading entry in column 1, which is to the *left* of the one above. The leading entries do not move to the right, so (c) is not even REF.

Step 5. (d) Row 1 leads in column 1, row 2 leads in column 3, row 3 is zero and at the bottom: REF. Leading entries are 1. Column 1 has no other nonzero entry; column 3 has only the 1 (the entry above it is 0; the 3 and 2 lie in non-pivot columns, which is allowed). So (d) is in RREF.

Answer: (a) RREF. (b) REF only: the entry 2 above the pivot in column 3 is nonzero. (c) Neither: leading entries do not move to the right. (d) RREF.

2. [Foundational]

Use elementary row operations to reduce $\begin{pmatrix}1&2&1\\2&5&3\\1&3&4\end{pmatrix}$ to RREF. Write each operation in the form $R_i \to R_i - cR_j$ and state the pivot columns.

Step 1. Strategy: take each column in turn from the left, make its pivot equal to 1, then clear every other entry in that column using the pivot row.

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix}1&2&1\\0&1&1\\0&1&3\end{pmatrix}$$

Step 3. Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 - R_2$:

$$\begin{pmatrix}1&0&-1\\0&1&1\\0&0&2\end{pmatrix}$$

Step 4. Apply $R_3 \to \frac{1}{2}R_3$:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 5. Apply $R_1 \to R_1 + R_3,\ R_2 \to R_2 - R_3$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 6. The matrix is now in RREF. It has a pivot in each of columns $1, 2, 3$.

Answer: RREF $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$; pivot columns $1, 2, 3$.

3. [Intermediate]

Reduce $\begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 0 & 2 \\ 1 & 2 & 1 & -1 \end{pmatrix}$ to RREF. State the pivot columns and explain why column 2 is not a pivot column.

Step 1. Work column by column. Column 1 already has a 1 in the top position, so use it to clear below.

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 2 & -4 \\ 0 & 0 & 2 & -4 \end{pmatrix}$$

Step 3. Apply $R_2 \to \frac{1}{2}R_2$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 2 & -4 \end{pmatrix}$$

Step 4. Apply $R_1 \to R_1 + R_2,\ R_3 \to R_3 - 2R_2$:

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Step 5. After clearing column 1, column 2 below row 1 is entirely 0, so there is no pivot in column 2 (nothing available to turn into a leading entry). Move on to column 3: it has a nonzero entry in row 2, which becomes the pivot.

Step 6. The matrix is in RREF with pivot columns 1 and 3.

Answer: RREF $= \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Pivot columns $1, 3$; column 2 has no pivot because, once column 1 is cleared, every entry of column 2 in rows 2 and 3 is 0.

4. [Intermediate]

Reduce $\begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 1 \\ 2 & 0 & -2 \end{pmatrix}$ to RREF. A row swap is needed: say why, and state the pivot columns.

Step 1. The entry in position $(1,1)$ is 0, so it cannot serve as a pivot. We must bring up a row whose first entry is nonzero: swap row 1 with row 2 (or row 3). The swap is an allowed elementary row operation and does not change the solution set of a system.

Step 2. Apply $R_1 \leftrightarrow R_2$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 2 & 0 & -2 \end{pmatrix}$$

Step 3. Apply $R_3 \to R_3 - 2R_1$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & -2 & -4 \end{pmatrix}$$

Step 4. Apply $R_2 \to \frac{1}{2}R_2$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{pmatrix}$$

Step 5. Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 + 2R_2$:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Step 6. The matrix is now in RREF.

Answer: RREF $= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$; pivot columns 1,2.

5. [Challenge]

Over $\mathbb{C}$, reduce $\begin{pmatrix} 1 & i & 0 \\ 2 & 2i & 1 \\ 1 & i & 1 \end{pmatrix}$ to RREF. State the pivot columns.

Step 1. Arithmetic over $\mathbb{C}$ follows the same rules; the only new fact needed is $i^2 = -1$. Column 1 has a 1 at the top, so clear below it. The multiples of i in column 2 will be cleared automatically along with column 1.

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix} 1 & i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 3. Apply $R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Step 4. Column 2 is zero below row 1 (it is i times column 1), so it holds no pivot. Column 3 gives the

second pivot.

Answer: RREF $= \begin{pmatrix} 1 & i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$; pivot columns $1, 3$.

6. [Challenge]

(Why row reduction works.) Let S' be obtained from a system S by one elementary row operation $R_i \to R_i + \lambda R_j$ $(i \neq j)$. Prove that S and S' have the same solutions.

Step 1. Let the equations of S be $E_1, \ldots, E_m$ (each equation is an expression E_k equal to a number β_k). S' has the same equations except that E_i is replaced by $E_i + \lambda E_j$ (and β_i by $\beta_i + \lambda\beta_j$).

Step 2. ($\Rightarrow$) Suppose $\mathbf{x}$ solves every equation of S. Then $E_i(\mathbf{x}) = \beta_i$ and $E_j(\mathbf{x}) = \beta_j$. Hence $(E_i + \lambda E_j)(\mathbf{x}) = \beta_i + \lambda\beta_j$, so $\mathbf{x}$ also satisfies the new equation. All other equations are unchanged, so $\mathbf{x}$ solves S'.

Step 3. ($\Leftarrow$) Suppose $\mathbf{x}$ solves S'. Then it satisfies $E_j = \beta_j$ (unchanged) and $E_i + \lambda E_j = \beta_i + \lambda\beta_j$. Subtract λ times the first from the second: $E_i(\mathbf{x}) = \beta_i$. So $\mathbf{x}$ solves S.

Step 4. Both directions hold, so the solution sets are equal. The same argument (or the observation that each operation can be undone by another operation of the same kind) covers swaps and scaling by $c \neq 0$.

Answer: Each solution of S solves S', and subtracting λE_j recovers E_i, so each solution of S' solves S; the solution sets are equal.

Part B — Solving Linear Systems

7. [Foundational]

Solve by row reduction: $\begin{cases} x + y = 3 \\ 2x - y = 0 \end{cases}$

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array}\right)$$

Step 3. Apply $R_2 \to \dfrac{1}{-3}R_2$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right)$$

Step 4. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right)$$

Step 5. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

Step 6. Read each nonzero row as an equation for its basic variable:

$x = 1$

$y = 2$

Step 7. Check by substituting $(x, y) = (1, 2)$ into the original equations: the left sides give $(3, 0)$, the right-hand sides ✓.

Answer: Unique solution: $(x, y) = (1, 2)$.

8. [Foundational]

Solve by row reduction: $\begin{cases} x + y + 2z = 5 \\ 2x - y + z = 4 \\ x + 2y - z = 3 \end{cases}$

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 5 \\ 2 & -1 & 1 & 4 \\ 1 & 2 & -1 & 3 \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 5 \\ 0 & -3 & -3 & -6 \\ 0 & 1 & -3 & -2 \end{array}\right)$$

Step 3. Apply $R_2 \to \dfrac{1}{-3}R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & -3 & -2 \end{array}\right)$$

Step 4. Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -4 & -4 \end{array}\right)$$

Step 5. Apply $R_3 \to \dfrac{1}{-4}R_3$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Step 6. Apply $R_1 \to R_1 - R_3,\ R_2 \to R_2 - R_3$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Step 7. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

Step 8. Read each nonzero row as an equation for its basic variable:

$x = 2$

$y = 1$

$z = 1$

Step 9. Check by substituting $(x, y, z) = (2, 1, 1)$ into the original equations: the left sides give $(5, 4, 3)$, the right-hand sides ✓.

Answer: Unique solution: $(x, y, z) = (2, 1, 1)$.

9. [Intermediate]

Solve: $\begin{cases} x + 2y - z = 4 \\ 2x + 5y + z = 9 \end{cases}$ Describe the solution set in parametric vector form.

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & 5 & 1 & 9 \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & 3 & 1 \end{array}\right)$$

Step 3. Apply $R_1 \to R_1 - 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & -7 & 2 \\ 0 & 1 & 3 & 1 \end{array}\right)$$

Step 4. Pivot columns are the columns of the basic variables: x, y. The remaining variable(s) z are *free*: call them $z = s$.

Step 5. Read each nonzero row as an equation for its basic variable:

$$x = 2 + 7s$$
$$y = 1 - 3s$$
$$z = s$$

Step 6. Write this as a vector: $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}$ (the parameters s may be any numbers).

Step 7. Check with $s = 1$: $\mathbf{x} = \begin{pmatrix} 9 \\ -2 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(4, 9)$ ✓.

Answer: Infinitely many solutions: $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}$.

10. [Intermediate]

Decide whether the system is consistent: $\begin{cases} x + 2y + z = 1 \\ 2x + 4y + 3z = 3 \\ x + 2y - z = 2 \end{cases}$

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & 1 \end{array}\right)$$

Step 3. Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 + 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right)$$

Step 4. Row 3 reads $0 = 1$, a contradiction. No numbers x, y, z can satisfy it, so the system is *inconsistent*.

Answer: No solution (inconsistent).

11. [Intermediate]

Solve $\begin{cases} x + y + z + w = 2 \\ x + 2y + 3w = 5 \end{cases}$ and identify the basic and free variables.

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 3 & 5 \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - R_1$:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & 2 & 3 \end{array}\right)$$

Step 3. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & -1 \\ 0 & 1 & -1 & 2 & 3 \end{array}\right)$$

Step 4. Pivot columns are the columns of the basic variables: x, y. The remaining variable(s) z, w are *free*: call them $z = s$, $w = t$.

Step 5. Read each nonzero row as an equation for its basic variable:

$$x = -1 - 2s + t$$
$$y = 3 + s - 2t$$
$$z = s$$
$$w = t$$

Step 6. Write this as a vector: $\mathbf{x} = \begin{pmatrix} -1 \\ 3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix}$ (the parameters s, t may be any numbers).

Step 7. Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -2 \\ 2 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(2, 5)$ ✓.

Answer: Infinitely many solutions: $\mathbf{x} = \begin{pmatrix} -1 \\ 3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix}$.

12. [Intermediate]

Solve the homogeneous system $\begin{cases} x + 2y - w = 0 \\ 2x + 4y + z + w = 0 \\ x + 2y + z + 2w = 0 \end{cases}$ and write the solution set as a span.

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 0 \\ 2 & 4 & 1 & 1 & 0 \\ 1 & 2 & 1 & 2 & 0 \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{array}\right)$$

Step 3. Apply $R_3 \to R_3 - R_2$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Step 4. Pivot columns are the columns of the basic variables: x, z. The remaining variable(s) y, w are *free*: call them $y = s$, $w = t$.

Step 5. Read each nonzero row as an equation for its basic variable:

$$\begin{aligned} x &= -2s + t \\ y &= s \\ z &= -3t \\ w &= t \end{aligned}$$

Step 6. Write this as a vector: $\mathbf{x} = s\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 0 \\ -3 \\ 1 \end{pmatrix}$ (the parameters s, t may be any numbers).

Step 7. Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -1 \\ 1 \\ -3 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0, 0)$ ✓.

Answer: Infinitely many solutions: $\mathbf{x} = s\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 0 \\ -3 \\ 1 \end{pmatrix}$. Equivalently the solution set is the span of the vectors multiplying the parameters (a subspace of $\mathbb{R}^4$; it always contains $\mathbf{0}$).

13. [Intermediate]

Over $\mathbb{C}$, solve $\begin{cases} x + iy = 1 + i \\ ix + 2y = i \end{cases}$

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ i & 2 & i \end{array}\right)$$

Step 2. Apply $R_2 \to R_2 - iR_1$:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ 0 & 3 & 1 \end{array}\right)$$

Step 3. Apply $R_2 \to \frac{1}{3}R_2$:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ 0 & 1 & \frac{1}{3} \end{array}\right)$$

Step 4. Apply $R_1 \to R_1 - iR_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 1+\frac{2}{3}i \\ 0 & 1 & \frac{1}{3} \end{array}\right)$$

Step 5. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

Step 6. Read each nonzero row as an equation for its basic variable:

$$x = 1 + \frac{2}{3}i$$
$$y = \frac{1}{3}$$

Step 7. Check by substituting $(x, y) = (1+\frac{2}{3}i, \frac{1}{3})$ into the original equations: the left sides give $(1+i, i)$, the right-hand sides ✓.

Answer: Unique solution: $(x, y) = (1 + \frac{2}{3}i, \frac{1}{3})$.

14. [Challenge]

For which real k does the system $\begin{cases} x + y + z = 1 \\ x + 2y + 3z = 2 \\ x + 3y + kz = 4 \end{cases}$ have (i) a unique solution, (ii) no solution, (iii) infinitely many solutions?

Step 1. Write the augmented matrix and reduce, keeping k as a symbol: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 1 & 3 & k & 4 \end{array}\right)$.

Step 2. Apply $R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1$: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & k-1 & 3 \end{array}\right)$.

Step 3. Apply $R_3 \to R_3 - 2R_2$: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k-5 & 1 \end{array}\right)$.

Step 4. Read the last row: $(k-5)z = 1$. If $k \neq 5$ we can divide by $k-5$, so $z = \dfrac{1}{k-5}$; then y and x follow by back-substitution. There are three pivots, so the solution is unique.

Step 5. If $k = 5$ the last row reads $0 = 1$, a contradiction: no solution.

Step 6. Infinitely many solutions would need a free variable and a consistent system. For $k = 5$ the system is inconsistent, and for $k \neq 5$ there is no free variable. So this never happens.

Step 7. Check one value: $k = 6$ gives $z = 1$, then $y + 2 = 1$ so $y = -1$, then $x - 1 + 1 = 1$ so $x = 1$; test in equation 3: $1 + 3(-1) + 6(1) = 4$ ✓.

Answer: (i) unique solution for every $k \neq 5$; (ii) no solution when $k = 5$; (iii) never infinitely many.

15. [Challenge]

A homogeneous system has m equations and n unknowns with $n > m$. Prove it has a nonzero solution.

Step 1. Reduce the $m \times n$ coefficient matrix to RREF. Each pivot occupies its own row and its own column, so there are at most m pivots (one per row).

Step 2. The coefficient matrix has n columns and $n > m \geq$ (number of pivots). So at least $n - m \geq 1$ columns contain no pivot; the variables in those columns are free.

Step 3. A homogeneous system is always consistent (the zero vector solves it). Give one free variable the value 1 (others 0) and solve for the basic variables from the RREF rows. The resulting solution has a coordinate equal to 1, so it is nonzero.

Answer: At most m pivot columns, so at least $n - m \geq 1$ free variables; setting a free variable to 1 gives a nonzero solution of the always-consistent homogeneous system.

Part C — Column Spaces, Relations and Proofs

16. [Foundational]

The matrix $R = \begin{pmatrix} 1 & 0 & 3 & 0 & -2 \\ 0 & 1 & -1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 5 \end{pmatrix}$ is the RREF of the coefficient matrix of a homogeneous system. State the pivot columns, the basic and free variables (use $x_1, \ldots, x_5$), and the general solution of $R\mathbf{x} = \mathbf{0}$.

Step 1. Pivot columns are the columns containing a leading 1: columns 1, 2 and 4. So x_1, x_2, x_4 are basic and x_3, x_5 are free; set $x_3 = s$, $x_5 = t$.

Step 2. Read the rows: row 1: $x_1 + 3x_3 - 2x_5 = 0$, so $x_1 = -3s + 2t$. Row 2: $x_2 - x_3 + 4x_5 = 0$, so $x_2 = s - 4t$. Row 3: $x_4 + 5x_5 = 0$, so $x_4 = -5t$.

Step 3. Vector form: $\mathbf{x} = \begin{pmatrix} -3s + 2t \\ s - 4t \\ s \\ -5t \\ t \end{pmatrix} = s \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -4 \\ 0 \\ -5 \\ 1 \end{pmatrix}$.

Step 4. Check the first vector in row 1: $(-3) + 3(1) - 2(0) = 0$ ✓; in row 2: $1 - 1 + 0 = 0$ ✓; in row 3: $0 + 0 = 0$ ✓.

Answer: Pivots: columns $1, 2, 4$. Free: x_3, x_5. $\mathbf{x} = s(-3, 1, 1, 0, 0)^T + t(2, -4, 0, -5, 1)^T$.

17. [Intermediate]

Let $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{pmatrix}$. (a) Reduce A to RREF. (b) Give a basis of the column space $\mathrm{col}(A)$ made of columns of A itself. (c) Express each non-pivot column of A as a linear combination of the pivot columns.

Step 1. (a) Row reduce.

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Step 3. Apply $R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Step 4. The RREF has pivots in columns 1 and 3.

Step 5. (b) Rule: the pivot columns of the *original* matrix A (not of the RREF) form a basis of $\mathrm{col}(A)$. So a basis is $\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Step 6. (c) The entries of a non-pivot column of the RREF are the coefficients: column 2 of the RREF is $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$, so $\mathrm{col}_2 = 2\,\mathrm{col}_1$; column 4 of the RREF is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so $\mathrm{col}_4 = 1\,\mathrm{col}_1 + 1\,\mathrm{col}_3$.

Step 7. Check column 4: $1 \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ ✓.

Answer: RREF $= \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Basis of $\mathrm{col}(A)$: $\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$ (columns 1 and 3 of A); $\mathrm{col}_2 = 2\,\mathrm{col}_1$, $\mathrm{col}_4 = \mathrm{col}_1 + \mathrm{col}_3$.

18. [Intermediate]

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$, reduce to RREF and give a basis of the row space $\mathrm{row}(A)$ and the number of pivots.

Step 1. Row operations replace each row by a combination of rows, and they can be reversed, so they do not change the span of the rows. Hence $\mathrm{row}(A) = \mathrm{row}(R)$ where R is the RREF.

Step 2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 3R_1$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 3. Apply $R_1 \to R_1 - 3R_2,\ R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Step 4. The nonzero rows of the RREF are linearly independent (each has a 1 in a pivot column where all the others have 0) and span the row space. So they form a basis.

Answer: RREF $= \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$; basis of row$(A)$: $(1,2,0)$, $(0,0,1)$; the rank (number of pivots) is 2.

19. [Intermediate]

Decide whether $\mathbf{w} = (3,2,1)$ lies in $\text{span}\{(1,0,1),(2,1,0),(0,1,-2)\} \subseteq \mathbb{R}^3$ by row reducing one augmented matrix. Give the coefficients if it does.

Step 1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & -2 & 1 \end{array}\right)$$

Step 2. (The columns are the three vectors; the right-hand side is $\mathbf{w}$.)

Step 3. Apply $R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & -2 & -2 & -2 \end{array}\right)$$

Step 4. Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 + 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 2 \end{array}\right)$$

Step 5. Row 3 reads $0 = 1$, a contradiction. No numbers a, b, c can satisfy it, so the system is *inconsistent*.

Answer: No: the system is inconsistent, so $\mathbf{w} \notin$ span.

20. [Intermediate]

Place the vectors $(1,2,1,0),\ (1,0,1,1),\ (2,1,0,1)$ of $\mathbb{R}^4$ as columns of a matrix. Use RREF to decide whether they are linearly independent.

Step 1. Put the vectors in the columns of A; independence means $A\mathbf{a} = \mathbf{0}$ has only the zero solution.

Step 2. $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. Reduce it.

Step 3. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & -2 & -3 \\ 0 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$$

Step 4. Apply $R_2 \to \dfrac{1}{-2} R_2$:

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & \frac{3}{2} \\ 0 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$$

Step 5. Apply $R_1 \to R_1 - R_2,\ R_4 \to R_4 - R_2$:

$$\begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{3}{2} \\ 0 & 0 & -2 \\ 0 & 0 & -\frac{1}{2} \end{pmatrix}$$

Step 6. Apply $R_3 \to \frac{1}{-2}R_3$:

$$\begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{3}{2} \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{2} \end{pmatrix}$$

Step 7. Apply $R_1 \to R_1 - \frac{1}{2}R_3,\ R_2 \to R_2 - \frac{3}{2}R_3,\ R_4 \to R_4 + \frac{1}{2}R_3$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Step 8. Every column holds a pivot, so there are no free variables: the only solution of $A\mathbf{a} = \mathbf{0}$ is $\mathbf{a} = \mathbf{0}$.

Answer: Linearly independent (a pivot in every column).

21. [Intermediate]

Show that $(1,0,1), (1,1,0), (0,1,1), (2,3,1)$ in $\mathbb{R}^3$ are linearly dependent, and use RREF to find an explicit relation among them.

Step 1. Four vectors in $\mathbb{R}^3$: the matrix with these columns is 3×4, so it cannot have a pivot in every column. Dependence is guaranteed; we now find a relation.

Step 2. $A = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 \end{pmatrix}$. Reduce it.

Step 3. Apply $R_3 \to R_3 - R_1$:

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & -1 & 1 & -1 \end{pmatrix}$$

Step 4. Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 + R_2$:

$$\begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \end{pmatrix}$$

Step 5. Apply $R_3 \to \frac{1}{2}R_3$:

$$\begin{pmatrix}1&0&-1&-1\\0&1&1&3\\0&0&1&1\end{pmatrix}$$

Step 6. Apply $R_1 \to R_1 + R_3,\ R_2 \to R_2 - R_3$:

$$\begin{pmatrix}1&0&0&0\\0&1&0&2\\0&0&1&1\end{pmatrix}$$

Step 7. Column 4 has no pivot, so a_4 is free: set $a_4 = 1$. Solving the RREF rows gives the vector $\begin{pmatrix}0\\-2\\-1\\1\end{pmatrix}$.

Step 8. So the relation is $(0)v_1+(-2)v_2+(-1)v_3+(1)v_4 = \mathbf{0}$. Check: $(0)\begin{pmatrix}1\\0\\1\end{pmatrix}+(-2)\begin{pmatrix}1\\1\\0\end{pmatrix}+(-1)\begin{pmatrix}0\\1\\1\end{pmatrix}+(1)\begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$ ✓.

Answer: Dependent: $(0)v_1 + (-2)v_2 + (-1)v_3 + (1)v_4 = \mathbf{0}$.

22. [Intermediate]

In $P_2(\mathbb{R})$, decide whether $q(x) = 2 + 3x + x^2 \in \operatorname{span}\{1 + x,\ x + x^2,\ 1 + x^2\}$. Set up the system, reduce, and conclude.

Step 1. We need a, b, c with $a(1 + x) + b(x + x^2) + c(1 + x^2) = 2 + 3x + x^2$. Expand and collect powers: $(a + c) + (a + b)x + (b + c)x^2$.

Step 2. A polynomial equality means equal coefficients: constant $a + c = 2$; x: $a + b = 3$; x^2: $b + c = 1$. In matrix form (columns are the coordinate vectors with respect to $1, x, x^2$).

Step 3. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c}1&0&1&2\\1&1&0&3\\0&1&1&1\end{array}\right)$$

Step 4. Apply $R_2 \to R_2 - R_1$:

$$\left(\begin{array}{ccc|c}1&0&1&2\\0&1&-1&1\\0&1&1&1\end{array}\right)$$

Step 5. Apply $R_3 \to R_3 - R_2$:

$$\left(\begin{array}{ccc|c}1&0&1&2\\0&1&-1&1\\0&0&2&0\end{array}\right)$$

Step 6. Apply $R_3 \to \frac{1}{2}R_3$:

$$\left(\begin{array}{ccc|c}1&0&1&2\\0&1&-1&1\\0&0&1&0\end{array}\right)$$

Step 7. Apply $R_1 \to R_1 - R_3,\ R_2 \to R_2 + R_3$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

Step 8. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

Step 9. Read each nonzero row as an equation for its basic variable:

$$a = 2$$
$$b = 1$$
$$c = 0$$

Step 10. Check by substituting $(a, b, c) = (2, 1, 0)$ into the original equations: the left sides give $(2, 3, 1)$, the right-hand sides ✓.

Answer: Yes, q is in the span; the coefficients are: $(a, b, c) = (2, 1, 0)$. That is, $q = 2(1 + x) + 1(x + x^2) + 0(1 + x^2)$.

23. [Challenge]

Let $A\mathbf{x} = \mathbf{b}$ be consistent and let $\mathbf{p}$ be one solution. Prove that the solution set is exactly $\{\mathbf{p}+\mathbf{h} :\ A\mathbf{h} = \mathbf{0}\}$.

Step 1. Two inclusions. ($\supseteq$) Let $A\mathbf{h} = \mathbf{0}$. Then $A(\mathbf{p}+\mathbf{h}) = A\mathbf{p} + A\mathbf{h} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, so $\mathbf{p}+\mathbf{h}$ is a solution.

Step 2. ($\subseteq$) Let $\mathbf{x}$ be any solution of $A\mathbf{x} = \mathbf{b}$. Put $\mathbf{h} = \mathbf{x} - \mathbf{p}$. Then $A\mathbf{h} = A\mathbf{x} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}$, and $\mathbf{x} = \mathbf{p} + \mathbf{h}$.

Step 3. Both inclusions hold, so the sets are equal. (This is why the general solution always has the shape "particular solution + solution of the homogeneous system".)

Answer: $A(\mathbf{p} + \mathbf{h}) = \mathbf{b}$ for each $\mathbf{h}$ in the null set, and any solution $\mathbf{x}$ equals $\mathbf{p} + (\mathbf{x} - \mathbf{p})$ with $A(\mathbf{x} - \mathbf{p}) = \mathbf{0}$.

24. [Challenge]

Find the condition on a, b, c for the system $\begin{cases} x + 2y - z = a \\ 2x + 3y + z = b \\ x + y + 2z = c \end{cases}$ to be consistent.

Step 1. Row reduce the augmented matrix, carrying a, b, c along: $\left(\begin{array}{ccc|c} 1 & 2 & -1 & a \\ 2 & 3 & 1 & b \\ 1 & 1 & 2 & c \end{array}\right)$.

Step 2. $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$: $\left(\begin{array}{ccc|c} 1 & 2 & -1 & a \\ 0 & -1 & 3 & b - 2a \\ 0 & -1 & 3 & c - a \end{array}\right)$.

Step 3. $R_3 \to R_3 - R_2$: $\left(\begin{array}{ccc|c} 1 & 2 & -1 & a \\ 0 & -1 & 3 & b - 2a \\ 0 & 0 & 0 & (c - a) - (b - 2a) \end{array}\right)$. The last right-hand side simplifies to $a - b + c$.

Step 4. The last row says $0 = a - b + c$. If $a - b + c \neq 0$ this is a contradiction (inconsistent). If $a - b + c = 0$ the row is $0 = 0$ and the system is consistent, with a free variable (infinitely many solutions).

Step 5. Check: take $(a, b, c) = (1, 3, 2)$, which satisfies $1 - 3 + 2 = 0$. Then $x = 1 - 2y + z$ from

equation 1; with $z = 0$: $x = 1 - 2y$, equation 2: $2 - 4y + 3y = 3$, so $y = -1$, $x = 3$; equation 3: $3 - 1 + 0 = 2$ ✓.

Answer: Consistent if and only if $a - b + c = 0$ (then infinitely many solutions); otherwise no solution.

25. [Challenge]

Let R be the RREF of A. Prove that every non-pivot column of R is a linear combination of the pivot columns of R, and explain why the same relation holds among the corresponding columns of A.

Step 1. Let the pivot columns of R be $j_1 < \cdots < j_r$. Column j_k of R is the standard vector $\mathbf{e}_k$ (a 1 in row k, zeros elsewhere), because each pivot is 1 and is the only nonzero entry in its column.

Step 2. Any column $\mathbf{c}$ of R has nonzero entries only in rows $1, \ldots, r$ (rows below the last pivot are zero rows). So $\mathbf{c} = c_1\mathbf{e}_1 + \cdots + c_r\mathbf{e}_r = c_1R_{j_1} + \cdots + c_rR_{j_r}$, where c_k is the k-th entry of $\mathbf{c}$.

Step 3. So a non-pivot column is the combination of pivot columns with its own entries as coefficients. Rewrite as a vector equation $\lambda_1R_1 + \cdots + \lambda_nR_n = \mathbf{0}$ (with a coefficient -1 on the non-pivot column): this is a solution of $R\boldsymbol{\lambda} = \mathbf{0}$.

Step 4. $A\boldsymbol{\lambda} = \mathbf{0}$ and $R\boldsymbol{\lambda} = \mathbf{0}$ have the same solutions, because row operations preserve solution sets (Problem 6). So the same coefficients give the same relation among the columns of A.

Answer: Each non-pivot column of R equals $\sum_k c_k\,\mathbf{e}_k$ with $\mathbf{e}_k$ a pivot column; the relation is a solution of $R\boldsymbol{\lambda} = \mathbf{0}$, hence of $A\boldsymbol{\lambda} = \mathbf{0}$.